HyperKD: Lifelong Hyperspectral Image Classification With Cross-Spectral–Spatial Knowledge Distillation

Zhenlin Li, Shaobo Xia, Jun Yue, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification models suffer from a phenomenon known as catastrophic forgetting, which refers to the sharp decline in performance on previously learned tasks after learning a new one when continuously acquiring new knowledge from a sequence of tasks. In recent years, some lifelong learning approaches have been proposed for HSI classification. Despite some progress, the challenge of catastrophic forgetting in lifelong learning remains significant and unresolved. In this article, we propose a novel lifelong learning framework for HSI classification, which is based on exemplar replay and cross-spectral–spatial feature knowledge distillation (KD), termed HyperKD. Specifically, the proposed framework incorporates a min–max cross-selection (MMCS) module tailored to HSI characteristics with a cross-spectral–spatial knowledge distillation (CSSKD) module. The MMCS module selects the most representative or diverse samples as exemplars from previous tasks for replay. Additionally, the CSSKD module not only transfers the prediction logit distribution from the previous network to the current network but also transfers the spectral–spatial feature distribution via cross-network KD, without directly assessing the similarity of feature distributions, thereby retaining more knowledge and mitigating forgetting. Through experiments conducted on a series of tasks, including the Pavia, Indian Pines, Salinas, and Houston datasets, our approach demonstrates superior performance compared to previous lifelong learning methods for HSI classification, effectively mitigating catastrophic forgetting. The code implementation of our approach will be publicly available at https://github.com/lzlsxs/hyperkd.**

***Index Terms*— Cross-spectral–spatial feature, exemplar replay, hyperspectral image (HSI) classification, incremental learning, knowledge distillation (KD), lifelong learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) are captured by spectral imagers across multiple contiguous spectral bands, resulting in each pixel of the image being characterized as a high-dimensional vector comprising tens to hundreds of spectral bands. Compared to other optical images [1], [2], [3], [4], [5], HSI classification is a pixel-level endeavor, entailing the categorization of each individual pixel. HSI classification has significant applications in various fields such as smart agriculture, mineral exploration, environmental monitoring, and military target reconnaissance [6], [7], [8], [9], [10]. Recently, various deep learning-based HSI classification methods have been proposed, yielding impressive classification outcomes [11], [12], [13], [14], [15], [16].

The rapid advancement of remote sensing technology enables researchers to continuously acquire a large number of HSIs. This trend requires neural networks to develop learning capabilities similar to those of the human brain, allowing them to continuously absorb new knowledge from images collected by various remote sensing satellites or the same satellite in different geographical regions, thereby accurately identifying new land-cover types. In practical applications, several factors, including limited storage space, strict data security requirements, and privacy policies, significantly constrain the storage of training data, often permitting the retention of only a small fraction or necessitating the discarding of all data. In these scenarios, neural networks can only access data from the current task while being evaluated on all previously learned tasks. This limitation makes them susceptible to catastrophic forgetting during continuous learning, where acquiring new information often results in the loss of previously learned knowledge, ultimately resulting in a decline in overall performance [17], [18].

To alleviate catastrophic forgetting, the concept of lifelong learning (also known as incremental learning or continuous learning) has been proposed [19], [20], [21]. Lifelong learning enables computer vision systems to continuously acquire new knowledge from new tasks while simultaneously retaining knowledge from the previously learned tasks [22], [23]. Lifelong learning can be categorized into three types based on learning scenarios [24]: task incremental learning (TIL), in which the model has access to the task identity during inference; domain incremental learning (DIL), characterized by constant task structure but varying input distribution; and class incremental learning (CIL), where the model does not have access to the task identity during inference and must, therefore, be able to distinguish between all classes from all tasks. This article focuses exclusively on CIL, with any

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Zhenlin Li is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China (e-mail: lzlsxs@hnu.edu.cn).

Shaobo Xia is with the Department of Geomatics Engineering, Changsha University of Science and Technology, Changsha 410114, China (e-mail: shaoboxia2020@gmail.com).

Jun Yue is with the School of Automation, Central South University, Changsha 410083, China (e-mail: junyue@csu.edu.cn).

Leyuan Fang is with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with Peng Cheng Laboratory, Shenzhen 518000, China (e-mail: fangleyuan@gmail.com).

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subsequent mention of lifelong learning implicitly referring to CIL unless otherwise noted.

In the context of lifelong learning based on neural networks, a common naive approach is to utilize finetuning with a sole cross-entropy loss to continually learn from new tasks. However, this lifelong learning approach learns each task in isolation, neglecting any data or knowledge from prior tasks. As a result, the trained model exhibits a bias toward new classes, resulting in the loss of classification capabilities for old classes. Based on finetuning, numerous research studies have concentrated on lifelong learning and proposed some methods. These methods can be broadly categorized into four types [25]: weight regularization methods, distillation-based methods, rehearsal-based methods, and bias correction methods. Weight regularization methods estimate the importance of each parameter in the network after each task and penalize changes to parameters according to their significance during the training of a new task. Representative methods include EWC [26], MAS [27], AFEC [28], SI [29], and AGS [30]. Distillation-based methods utilize the knowledge distillation (KD) technique to direct the current network to mimic the intermediate feature distribution or the prediction logit distribution of previous networks, thereby preserving prior knowledge and mitigating catastrophic forgetting. Representative methods include LWF [31], LFL [32], LWM [33], DMC [34], ANCL [25], and FCS [35]. Rehearsal-based methods address catastrophic forgetting by retaining exemplars from past tasks or generating synthetic samples through generative models such as diffusion models [36]. These mechanisms ensure that partial information of prior task distributions is replayed during new task training. Representative methods include ICaRL [23], EEIL [37], DGR [38], CAN [39], DGM [40], and FearNet [41]. Bias correction methods address task-recency bias where the network is biased toward the data of new tasks, potentially leading to distorted predictions of the model. Representative methods include BiC [42], IL2M [43], LUCIR [44], WA [45], and PODNet [46].

Lifelong learning has also been applied to HSI classification. In recent years, several lifelong learning approaches for HSI classification based on the previously mentioned methods have been proposed [47], [48], [49], [50], [51]. Xu et al. [47] utilized the herding sampling strategy [23] to select exemplars from previous task data and integrated the cross-entropy loss function with the distillation loss function to formulate the ultimate loss function. Essentially, this approach constitutes LWF augmented with example replay. Zhao et al. [48] proposed a decoupled structure consisting of a feature extractor and a classifier. In the initial stage, the feature extractor is trained, and its parameters are then frozen for all subsequent tasks. For each subsequent task, only the classifier is updated. During the classification stage, the distance between the test samples and the prototypes is calculated to perform classification. Zhao et al. [49] proposed a lifelong learning approach named CSSFD, specifically designed for HSIs. It incorporates a dedicated backbone architecture and utilizes cosine similarity to distill the spectral–spatial features. Additionally, it utilizes the hard distillation technique to distill the prediction logits of the model. Bai et al. [50] proposed GS2FIN-CIL, a lifelong learning framework that designs a dedicated feature extraction network to extract spectral–spatial features. During distillation, an angle loss function is incorporated into the overall loss function to mitigate catastrophic forgetting. Yu et al. [51] proposed DCPRN, a framework that employs prototype learning to derive a feature representation for each category in the previous task. The new task utilizes the feature representation obtained from the previous task and the data from the current task for training and KD, thereby preserving learned knowledge from the previous task. Despite advancements in lifelong learning for HSI classification, the challenge of catastrophic forgetting remains prominent and unresolved, highlighting the need for further optimization in the following key areas.

1) Exemplars play a crucial role in lifelong learning, significantly enhancing the robustness and performance of classifiers [23]. Similarly, selecting the optimal exemplars in HSI lifelong learning is crucial for the quality of HSI classification. Nevertheless, there exists a scarcity of research focusing on exemplar selection strategies that are specifically designed for HSIs.
2) Spectral–spatial features are essential for HSI classification [52]. Some lifelong learning methods for HSI classification utilize KD to transfer the spectral–spatial feature distribution from the previous model to the current. These methods evaluate the similarity of the spectral–spatial feature distributions using mean squared error (mse), cosine similarity, or Kullback–Leibler (KL) divergence [49], [50], [51]. Given the intricate nature of the spectral–spatial feature distributions, we contend that these methods fail to accurately assess their similarity.
3) Since several existing lifelong learning approaches for HSI classification have tailored specialized backbones [49], [51], they suffer from limited universality and generalizability.

This article proposes HyperKD, a lifelong learning framework for HSI classification, to address and optimize the aforementioned issues. HyperKD integrates a min–max cross-selection (MMCS) module tailored to HSI characteristics and a cross-spectral-spatial KD (CSSKD) module. The key contributions of this study are threefold.

1) HyperKD incorporates a CSSKD module, enabling the transfer of spectral–spatial feature distributions via cross-network KD from the previous model to the current one for retaining acquired knowledge, without directly assessing the similarity of these distributions.
2) In the HyperKD framework, we propose an MMCS module for HSIs. This module first establishes a novel distance, termed spectral–spatial Gaussian distance (SSGD), to evaluate the similarity between samples. Subsequently, utilizing SSGD, the method selects the most representative or diverse samples as exemplars, which are stored in a memory buffer for replay in future tasks.
3) We conduct lifelong learning experiments on four HSI classification tasks, each sourced from a distinct HSI

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dataset. The experimental results, analyzed both quantitatively and qualitatively, demonstrate that our proposed framework significantly enhances classification performance and effectively mitigates catastrophic forgetting, outperforming previous methods.

The subsequent sections of this article are structured as follows. Section II introduces some recent research works on exemplar selection and KD for lifelong learning. Section III provides a comprehensive introduction to our novel lifelong learning framework, HyperKD, specifically designed for HSI classification. Section IV introduces the datasets, experimental setup, experimental results, and related analysis. Section V summarizes the work presented in this article.

II. Related Work

A. Exemplar Selection Strategy

In lifelong learning, representative samples for each class can be selected as examples from the previous tasks and stored in a memory buffer. During the new task, the stored exemplars are combined with the current task's data for training, ensuring that at least some data distributions from previous tasks are incorporated into the training process. These exemplars play a pivotal role in lifelong learning, even serving as a key factor in determining the robust classification performance [23].

Rebuffi et al. [23] proposed a novel exemplar selection approach named Herding, which involves projecting all samples in each category onto the feature space constructed by the model. In the feature space, for each category, Herding selects the samples closest to the mean sample as exemplars and stores them for future tasks. Chaudhry et al. [53] proposed three exemplar selection methods: random sampling (Random), plane distance-based sampling (Distance), and entropy-based sampling (Entropy). Random sampling involves randomly selecting samples. Plane distance-based sampling hypothesizes that samples closer to the decision boundary are more representative of the overall samples. Consequently, it computes the distances between each sample and the decision boundary, subsequently selecting the samples with the shortest distances, or equivalently, the lowest predicted probabilities, as exemplars. Entropy-based sampling posits that samples with high prediction uncertainty are more representative. It computes the entropy of all samples in each class based on the model's output and selects the samples with the highest entropy as exemplars. Bang et al. [54] proposed the diversity-aware memory update (DAMU) method, which sorts samples based on their uncertainty and selects samples at medium intervals from the sorted list.

In recent years, the exemplar selection strategy has also been applied to lifelong learning for HSI classification. For example, Xu et al. [47] directly employed the Herding method to select exemplars from previous HSI datasets.

B. KD for Lifelong Learning

KD was proposed to transfer knowledge from a large, complex teacher model to a small, lightweight student model. This process enables the student model to attain or closely approximate the performance and accuracy of the teacher model, effectively achieving model compression. KD can be categorized into three types: response-based KD, feature-based KD, and relation-based KD [49], [55]. Hinton et al. [56] proposed that the prediction logit distribution of the student model should mimic that of the teacher model to facilitate knowledge transfer from the teacher to the student. Romero et al. [57] proposed a method to enhance the performance of the student model by distilling the intermediate representations of the hidden layers from the teacher to the student network. Park et al. [58] proposed a relational KD approach that transfers mutual relations of data with distance-wise and angle-wise distillation losses from the teacher model to the student model. Wang et al. [59] proposed a distillation method called CrossKD, which delivers the intermediate features of the student's detection head to the teacher's detection head for object detection.

Some lifelong learning methods utilize the KD technique to guide the current network to mimic the intermediate feature distribution or predict the logit distribution of the previous network, thereby preserving prior knowledge and alleviating catastrophic forgetting. Representative methods in this field include LWF [31], LFL [32], LWM [33], DMC [34], ANCL [25], and FCS [35]. LWF proposes the distillation framework for lifelong learning by simply combining the prediction logit distillation loss and the cross-entropy loss. LFL preserves previously learned knowledge by freezing the weights of the softmax layer and distilling the feature maps of the last hidden layer from the old model to the new model. LWM utilizes attention distillation loss to transfer knowledge from the old network to the new network. DMC and ANCL, respectively, propose lifelong learning methods based on KD, leveraging auxiliary networks to transfer knowledge from both the previous and auxiliary networks to the new network, thereby enhancing classifiers' performance. FCS initially employs a feature calibration network to adapt the drift of the intermediate feature caused by model evolution, followed by the application of KD.

In recent years, several lifelong learning approaches based on KD have been applied to HSI classification. Based on this, innovations tailored to the characteristics of HSI have been made, yielding certain achievements [47], [49], [50], [51]. Xu et al. [47] proposed a lifelong learning approach for HSI classification that utilizes the KD technique to align the prediction logit distribution of the previous network and current network, thereby retaining the previously acquired knowledge. Zhao et al. [49] proposed a lifelong learning approach that not only mimics the prediction logit distribution but also the intermediate feature distribution of the previous network, thereby transferring the acquired spectral–spatial feature distribution to the current network. Bai et al. [50] proposed an angle loss during the KD process, which requires no correspondence between the input vector and the label vector. Instead, it is sufficient to manipulate the previous model directly to extract the bias toward the old classes, thereby preserving the historical knowledge. Yu et al. [51] proposed DCPRN, which utilizes a dual-KD module to implement feature-level and decision-level knowledge transfer simultaneously.

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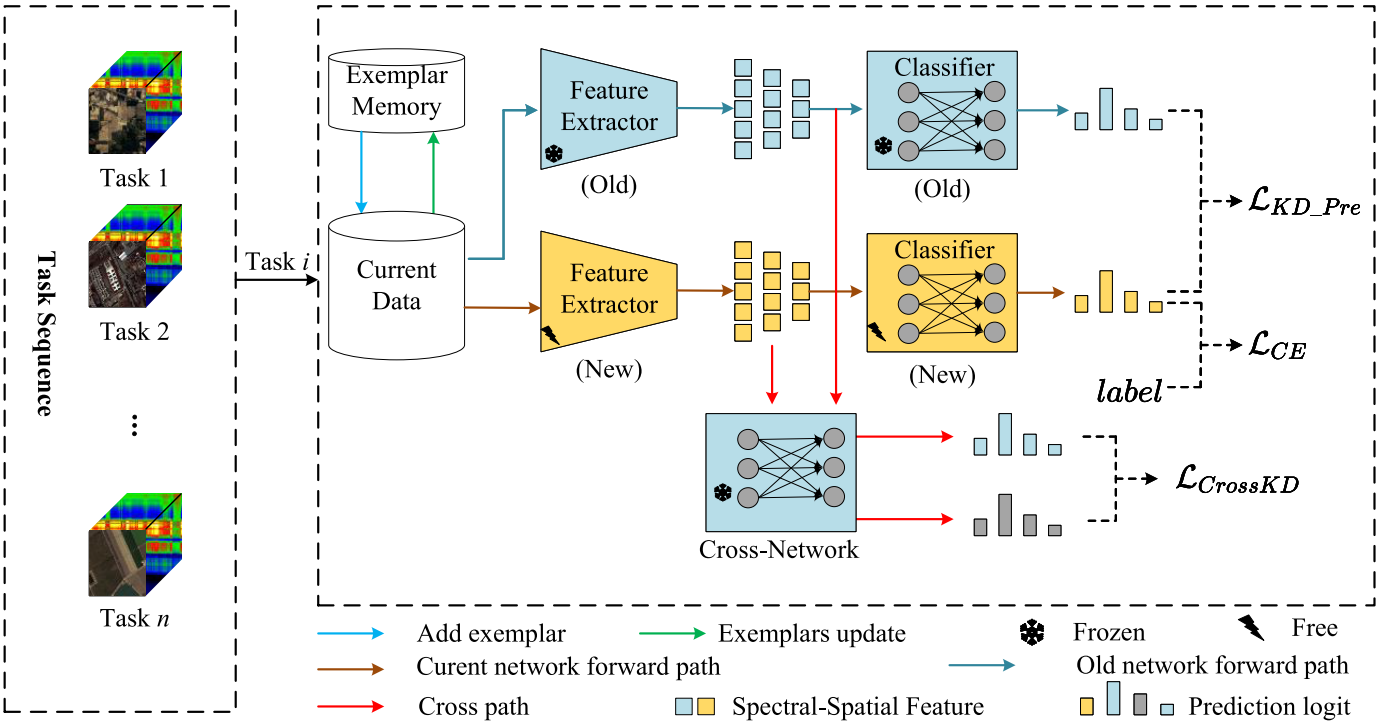


Fig. 1. Proposed lifelong learning framework HyperKD, integrating the MMCS module and the CSSKD module. MMCS selects the most representative or diverse samples as exemplars to update exemplar memory for replay. The CSSKD module primarily comprises the cross-entropy loss, the prediction logit distillation loss, and the cross-spectral–spatial feature distillation loss. The cross-spectral–spatial feature distillation loss employs a cross-network, a replica of the old classifier, to project the spectral–spatial features onto prediction logits. Subsequently, KD is utilized to direct the current network to mimic the spectral–spatial feature distribution of the old network.

III. Proposed Method

We present a detailed introduction to our novel lifelong learning framework, HyperKD, which is specifically designed for HSI classification. The HyperKD framework integrates the MMCS module, which selects the most representative or diverse samples as exemplars from previous tasks for replay, and the CSSKD module, which comprises the cross-entropy loss, the prediction logit distillation loss, and the cross-spectral–spatial feature distillation loss. The overall framework of the HyperKD is shown in Fig. 1. Section III-A introduces the problem setup for lifelong learning. Section III-B describes the MMCS module. Section III-C provides comprehensive details on the implementation of the CSSKD module.

A. Problem Setup

In lifelong learning, models are sequentially trained on a data stream $\mathcal{D} = \{D_t\}_{t=1}^{T}$, which consists of a sequence of T tasks. Each task's dataset $D_t = \{X_t, Y_t\}$ consists of input samples $X_t = \{x_{i,t}\}_{i=1}^{n_t}$ and their corresponding ground-truth labels $Y_t = \{y_{i,t} \in \mathcal{C}_t\}_{i=1}^{n_t}$, where n_t is the number of all samples in task t, $x_{i,t}$ denotes the ith sample, $\mathcal{C}_t$ represents the labels set, and $|\mathcal{C}_t|$ is the number of the classes in task t. $\mathcal{C}_t$ is the new classes in the task t, whereas they become the old classes in subsequent tasks. To be noted, the classes of different tasks must not overlap. In other words, $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$ if $i \neq j$. During the training of task t, models only access the training dataset $D_t = \{X_t, Y_t\}$ from task t (allowing for the storage of a small portion of data from previous tasks) and are evaluated on all seen classes $\cup_{i=1}^{t} Y_i$ from tasks 1 to t. This limitation results in catastrophic forgetting.

An ideal lifelong learning framework should demonstrate exceptional classification performance across all seen classes $\cup_{i=1}^{t} Y_i$ in tasks 1–t, even after being trained solely on the training dataset $D_t = \{X_t, Y_t\}$ from the new task t. In other words, it should effectively acquire new knowledge while retaining previously learned knowledge, similar to the human brain's ability. In this study, each task uses a distinct HSI dataset. We sequentially order the labels across all datasets for all tasks. Across different datasets, despite overlapping class names, each represents a different class and utilizes a different label for classification.

B. Min–Max Cross-Selection

When selecting exemplars, two key criteria should be met. First, samples selected from previous tasks for training the new task should represent their respective classes while exhibiting diversity. Second, selecting exemplars based on the model's feature layer or predicted output is discouraged, as in lifelong learning scenarios with long task sequences, the model undergoes continuous evolution, leading to a certain degree of randomness in the selecting process. To achieve the aforementioned objectives, we propose a novel exemplar selection method specifically designed for HSIs, termed MMCS. This method initially computes the distances between all samples and the mean sample using SSGD, subsequently selecting the samples that are proximal or distal to the mean as exemplars

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Algorithm 1 Exemplar Selection With MMCS

Input: $X = (X^s, \ldots, X^t)$ //new class data
Input: $P = (P^1, \ldots, P^{s-1})$ //stored exemplars
for $i = s$ to t **do**
  $x_i^{mean} \leftarrow \frac{1}{|X^i|} \sum_{x \in X^i} x$ //mean sample
  $P^i = [\,]$ // used to store exemplars
  for $j = 1$ to k **do** // select k exemplars for per class
    for $x \in X^i$ **do**
      if $x \in P^i$
        continue
      end if
      $Dis(x, x_i^{mean}) \leftarrow$ Calculate SSGD with (2)
      if is an even number **do**
        if $Dis(x, x_i^{mean}) < D_{min}$ **do**
          $D_{min} = Dis(x, x_i^{mean})$
          $x_{sel} = x$
        end if
      else
        if $Dis(x, x_i^{mean}) > D_{max}$ **do**
          $D_{max} = Dis(x, x_i^{mean})$
          $x_{sel} = x$
        end if
      end if
    end if
    $P^i \leftarrow P^i \cap x_{sel}$
  end if
  $P \leftarrow P^i \cap P$
end if
Output: P

through a cross-selection process. MMCS is presented in Algorithm 1.

1) Spectral–Spatial Gaussian Distance: HSI classification operates at the pixel level. How can the distance between two samples be effectively measured? Two intuitive methods can be considered. One method involves directly calculating the pixel-wise distance between two samples. However, this method relies solely on spectral information, neglecting spatial context. Alternatively, sampling can be centered on pixels to form a patch for each sample, followed by the calculation of patch-wise distances. Nevertheless, this direct calculation of the average distance between pixels within the patch fails to consider the varying weights of pixels at different locations within the patch.

To address the aforementioned issues, we propose the SSGD. When calculating the distance, the weight of each pixel at different locations within the sample patch is defined as follows:

$$G(i, j) = e^{-\frac{i^2+j^2}{2\sigma^2}} \tag{1}$$

where σ is the standard deviation, (i, j) denotes the spatial position coordinates, and the central pixel position is designated as (0, 0). Next, SSGD can be formulated as follows:

$$\mathrm{SSGD}(x, y) = \sum_{i=-m}^{m} \sum_{j=-n}^{n} G(i, j) * \|x_{i,j} - y_{i,j}\|_2 \tag{2}$$

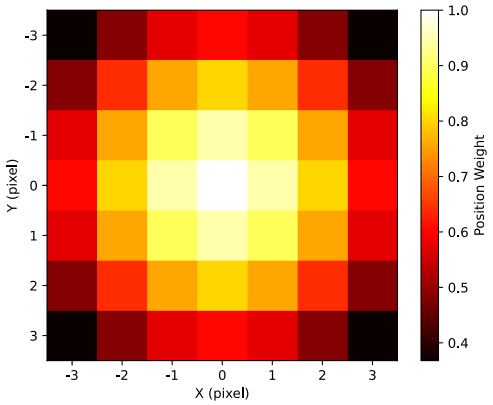


Fig. 2. Weight distribution map of pixel positions in a patch when the standard deviation σ is 3. Pixels located near the central pixel play a more significant role.

where x and y represent two sample patches, (i, j) denotes the spatial position coordinates, and m and n represent the maximum spatial coordinates along the two axes within each patch. Fig. 2 shows the schematic of different weights corresponding to different positions within the patch when $\sigma = 3$. Notably, during the computation of HSI sample distances by SSGD, the spectral information of the center pixel plays a more critical role than the surrounding pixels.

2) Min–Max Cross-Selection: For each category, we define a virtual mean sample to represent the collective characteristics. This virtual mean sample can be formulated as follows:

$$x_{\mathrm{mean}} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{3}$$

where n denotes the number of all samples for one category. We utilize (2) and (3) to compute the distances between each sample and the mean sample and then sort these distances. Subsequently, we cross-select K samples as exemplars based on either the minimum or maximum distance. The closest samples represent the majority of samples situated close to the mean, reflecting sample stability and enabling the trained model to generalize well to most samples. Conversely, the farthest samples exhibit diversity within the sample set, thereby broadening the decision boundary of the model. The illustration of the closest samples and farthest samples selected by MMCS is displayed in Fig. 3.

C. Cross-Spectral–Spatial KD

In this study, we propose a KD-based CSSKD module, specifically designed for HSIs, to facilitate lifelong learning and mitigate catastrophic forgetting. As depicted in Fig. 1, the network architecture encompasses a feature extractor F_θ and a classifier G_ϕ. For a new task, the network acquires training data alongside stored exemplars from previous tasks. Subsequently, these data are combined and fed into both the previous and current networks, respectively. Through the utilization of our proposed CSSKD module for training and optimization, a newly trained model is obtained. The CSSKD module primarily consists of the following three components.

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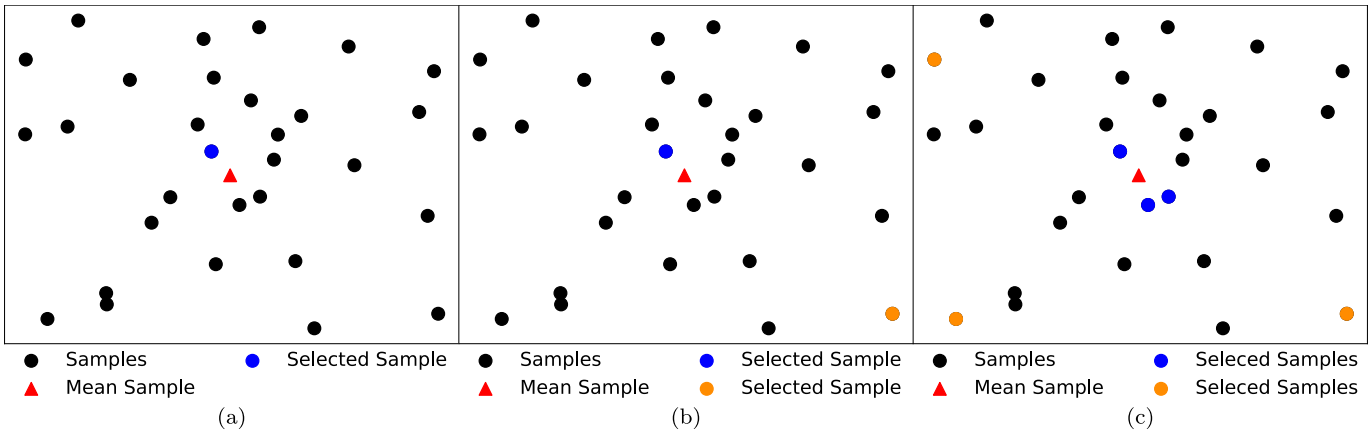


Fig. 3. Diagram of the selection of examples by MMCS. (a) Select a sample that is closest to the mean sample as an exemplar, marked by the blue dot. (b) Cross-select a sample that is farthest from the mean sample, marked by the orange dot. (c) Cross-select multiple samples as exemplars, marked by the blue and orange dots.

1) Cross-Entropy Loss: The cross-entropy loss function is commonly used to quantify the discrepancy between the output probability distribution of the model and the ground truth in multiclass classification tasks. In lifelong learning, the cross-entropy loss function is primarily used to learn the classification ability for new categories in new tasks, while simultaneously preserving the classification ability for previously learned categories through rehearsal using stored examples from previous tasks. The cross-entropy loss can be formulated as follows:

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{S} y_{\mathrm{ic}} \log\left(\frac{\exp\left(o_c\right)}{\sum_{j=1}^{S}\exp\left(o_j\right)}\right) \tag{4}$$

where N represents the total number of training samples, y_{ic} denotes the label assigned to each sample, S indicates the cumulative sum of all categories up to this point, and o represents the prediction logits generated by the network.

2) Prediction Logit Distillation Loss: This loss function utilizes KD, enabling the new network to mimic the prediction logit distribution of the previous network, thereby effectively transferring the class probability distribution knowledge from the old model to the new one, ensuring the preservation of existing knowledge. It can be formulated as follows:

$$L_{\mathrm{KD_Pre}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{S} p_{\mathrm{ic}}^{t-1}\log\left(p_{\mathrm{ic}}^{t}\right) \tag{5}$$

$$p_{\mathrm{ic}} = \frac{\exp\left(o_c/T\right)}{\sum_{j=1}^{S}\exp\left(o_j/T\right)} \tag{6}$$

where N represents the total number of training samples, S indicates the cumulative sum of all categories up to this point, o represents the output logits of the network, t and $t-1$ denote the current task and previous task, and T is the temperature scaling parameter.

3) Cross-Spectral–Spatial Feature Distillation Loss: Conventional methods for distilling spectral–spatial features rely on metrics such as mse, cosine similarity, or KL divergence to assess the similarity between feature distributions. Given the intricate nature of the spectral–spatial feature distributions, directly quantifying their similarity using metrics such as mse, cosine similarity, or KL divergence is challenging. We believe that measuring the similarity between the prediction logit distributions is more straightforward than measuring the similarity between the intricate spectral–spatial feature distributions. To address the challenges in measuring the similarity between the spectral–spatial feature distributions, we propose cross-spectral–spatial feature distillation loss that does not directly measure the similarity of these feature distributions. Instead, it employs a cross-network, a replica of the old classifier, to project the spectral–spatial features extracted by both the current and previous networks' feature extractors onto their respective prediction logits. Subsequently, the two prediction logits from the cross-network are distilled, ensuring that the spectral–spatial feature distributions of the current and previous networks are highly similar, enabling the transfer of these distributions from the previous to the current network, thereby preserving the previously acquired knowledge. This loss can be formulated as follows:

$$L_{\mathrm{CrossKD}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{S} p_{\mathrm{ic}}^{t-1}\log\left(\hat{p}_{\mathrm{ic}}^{t-1}\right) \tag{7}$$

where N represents the total number of training samples, S indicates the cumulative sum of all categories up to this point, p_{ic}^{t-1} denotes the output probabilities of the cross-network fed by the features extracted by the previous network's feature extractors, and $\hat{p}_{\mathrm{ic}}^{t-1}$ represents the output probabilities of the cross-network fed by the features extracted by the current network's feature extractors. Comparisons between conventional methods and our method for distilling spectral–spatial features are shown in Fig. 4.

Finally, the loss function in CSSKD is composed of the cross-entropy loss, the prediction logit distillation loss, and the cross-spectral–spatial feature distillation loss. It is formulated as follows:

$$L_{\mathrm{loss}} = L_{\mathrm{CE}} + \alpha L_{\mathrm{KD_Pre}} + \beta L_{\mathrm{CrossKD}} \tag{8}$$

where α and β are the coefficients. Detailed training steps of CSSKD are summarized in Algorithm 2.

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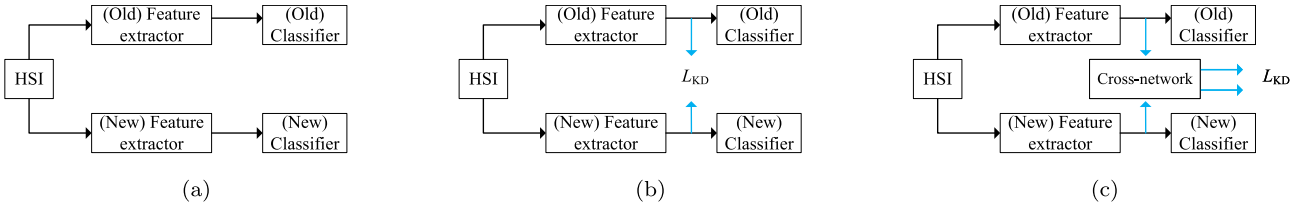


Fig. 4. Comparisons between conventional methods and our method for distilling spectral–spatial features. (a) Without spectral–spatial feature distillation. (b) Conventional spectral–spatial feature distillation. (c) Cross-spectral–spatial distillation. Cross-network is a replica of the old classifier.

Algorithm 2 Algorithm of CSSKD

Input: $X = (X^s, \ldots, X^t)$ //new task data
Input: $P = (P^1, \ldots, P^{s-1})$ //stored exemplars
Input: $\theta_{old}, \theta_{new}$ //old and new feature extractors
Input: ϕ_{old}, ϕ_{new} //old and new classifiers
$D \leftarrow X \cup P$
for $n = 1$ to $Epoch$ **do**
  for $(x, y) \in D$ **do**
    $f_{old} \leftarrow \theta_{old}(x)$ //feature from old network
    $f_{new} \leftarrow \theta_{new}(x)$ //feature from new network
    $p_{old} \leftarrow \phi_{old}(f_{old})$ //old network prediction
    $p_{new} \leftarrow \phi_{new}(f_{new})$ //new network prediction
    $\phi_{cross} \leftarrow \phi_{old}$ //cross-network
    $\hat{p} \leftarrow \phi_{cross}(f_{new})$ //cross prediction
    $L_{\mathrm{CE}}(y, p_{new}) \leftarrow$ Calculate the loss with (4)
    $L_{KD_Pre}(p_{old}, p_{new}) \leftarrow$ Calculate the loss with (5)
    $L_{KD_CSS}(p_{old}, \hat{p}) \leftarrow$ Calculate the loss with (7)
    $L_{\mathrm{loss}} = L_{\mathrm{CE}} + \alpha L_{KD_Pre} + \beta L_{KD_CSS}$ //the final loss
    $\theta, \phi \leftarrow -\nabla_{\theta,\phi}(L_{\mathrm{loss}})$
  end if
end if
Output: θ, ϕ

IV. Experiment and Analysis

A. Datasets Description

1) Pavia University Dataset: The Pavia University dataset is a hyperspectral dataset acquired by the reflective optics system imaging spectrometer (ROSIS-3) in 2003, covering the city of Pavia, Italy. It consists of 610 × 340 pixels, with a ground sampling distance of 1.3 m/pixel. Initially, 115 spectral bands were acquired; however, 12 were discarded due to noise contamination, leaving 103 bands available for practical applications. A total of 43 923 pixels were labeled, encompassing nine distinct land-cover classes.

2) Indian Pines Dataset: This hyperspectral dataset is frequently utilized for HSI classification. It was acquired in 1992 using the airborne visible infrared imaging spectrometer (AVIRIS) over a pine forest located in IN, USA. The dataset comprises 145 × 145 pixels, with a ground sampling distance of 20 m/pixel. Typically, bands 104–108, 150–163, and 220 are excluded from the original 220 spectral bands due to water absorption and inability to reflect light, resulting in a total of 200 bands available for application. A total of 10 366 pixels are labeled, encompassing 16 distinct land-cover classes.

3) Salinas Scene Dataset: The dataset was acquired by imaging the Salinas Valley in CA, USA, in 1992 using an airborne visible/infrared imaging spectrometer (AVIRIS). It features a spatial resolution of 512 × 217 pixels and a ground sampling distance of 3.7 m/pixel. The original dataset includes 224 spectral bands; however, water absorption renders bands 108–112, 154–167, and 224 ineffective for reflection. Consequently, the analysis typically utilizes bands 1–107, 113–153, and 168–223, resulting in a total of 204 bands. This dataset contains annotations for 154 129 pixel points, representing 16 distinct land-cover classes.

4) Houston Dataset: This dataset was acquired by the National Center for Airborne Laser Mapping (NCALM), funded by the National Science Foundation of the United States, using the ITRES CASI-1500 sensor on the campus of the University of Houston and nearby urban areas. It consists of 349 × 1905 pixels, with a ground sampling distance of 2.5 m for each pixel. The dataset covers 144 bands within a spectral range from 364 to 1046 nm. Among all the pixels, 15 029 pixels are labeled, which correspond to 15 different types of land cover.

The number of training and testing samples for each class in the four HSI datasets mentioned above is presented in Table I.

B. Experimental Setting

1) Task Setting: In our experiment, we design four tasks for sequential learning, each task utilizing a distinct HSI dataset. Task 1 utilizes the Pavia University dataset, task 2 the Indian Pines dataset, task 3 the Salinas Scene dataset, and task 4 the Houston dataset. During the training phase, we select and store five samples as exemplars from each class in the previous tasks for subsequent training of new tasks. During the testing phase, we are unaware of the specific task identity for each test sample.

2) Hardware Configuration: All experiments are conducted on a computing platform equipped with an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory, an Intel Xeon Platinum 8260C CPU (2.30 GHz), 86 GB of RAM, Ubuntu 20.04 OS, and the following software: PyTorch 2.1.1, Torchvision 0.16.1, Numpy 1.25.2, Python 3.10.12, CUDA 11.8, and cuDNN 8.

3) Parameter Setting: Four HSI datasets have spectral dimensions of 103, 220, 224, and 144 bands, respectively. Initially, 36 principal components are extracted for each dataset using principal component analysis (PCA). Subsequently, we perform sampling centered on each pixel in the image, using a patch size of 7, resulting in a 7 × 7 × 36 patch for each pixel. We use LSGA [11] as the backbone network. During the training phase, we utilize Adam as the optimizer,

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES PER CLASS ACROSS FOUR DATASETS

Pavia			Indian Pines			Salinas			Houston		
Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
Asphalt	548	6304	Corn Notill	50	1384	Brocoli Green Weeds1	200	1809	Healthy Grass	198	1053
Meadows	540	18146	Corn Mintill	50	784	Brocoli Green Weeds2	200	3526	Stressed Grass	190	1064
Gravel	392	1815	Corn	50	184	Fallow	200	1776	Synthetic Grass	192	505
Trees	524	2912	Grass Pasture	50	447	Fallow Rough Plow	200	1194	Tree	188	1056
Metal Sheets	265	1113	Grass Tree	50	697	Fallow Smooth	200	2478	Soil	186	1056
Bare Soil	532	4572	Hay Windrowed	50	439	Stubble	200	3759	Water	182	143
Bitumen	375	981	Soybean Notill	50	918	Celery	200	3379	Residential	196	1072
Bricks	514	3364	Soybean Mintill	50	2418	Grapes Untrained	200	11071	Commercial	191	1053
Shadows	231	795	Soybean clean	50	564	Soil Vinyard Develop	200	6003	Road	193	1059
			Wheat	50	162	Corn Senesced Green Weeds	200	3078	Highway	191	1036
			Woods	50	1244	Lettuce Romaine 4wk	200	868	Railway	181	1054
			Build Grass Tree Drives	50	330	Lettuce Romaine 5wk	200	1727	Parking Lot1	192	1041
			Stone Steel Towers	50	45	Lettuce Romaine 6wk	200	716	Parking Lot2	184	285
			Alfalfa	15	39	Lettuce Romaine 7wk	200	870	Tennis Court	181	247
			Grass Pasture Mowed	15	11	Vinyard Untrained	200	7068	Running Track	187	473
			Oats	15	5	Vinyard Vertical Trellis	200	1607			
Total	3921	40002	Total	695	9671	Total	3200	50929	Total	2832	12197

with a batch size of 128 and a learning rate of $5e^{-4}$. Both weight decay and momentum are set to 0, and the number of iterations (epochs) is 150. For the two coefficient parameters α and β in (9), we set them to 1 and 0.95, respectively.

4) Evaluation Metric: In the classification task, we assess the performance of the classifier across categories or datasets using three key metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient. In the context of lifelong learning, we propose task accuracy (TAC), which quantifies the performance of the classifier after each task. Additionally, we define average TAC (Ave), a metric that evaluates the average classification capability of the classifier at different stages after all tasks have been trained. Collectively, these metrics offer a comprehensive assessment of the classifier's learning and memory capabilities across diverse datasets and tasks. TAC and Ave can be formulated as follows:

$$\mathrm{TAC}(j) = \frac{1}{j}\sum_{i=1}^{j} \mathrm{acc}_{i,j} \tag{9}$$

$$\mathrm{Ave} = \frac{1}{T}\sum_{j=1}^{T} \mathrm{TAC}(j) \tag{10}$$

where $\mathrm{acc}_{i,j}$ denotes the test accuracy of the model on the testing dataset of task i after training on task j, $\mathrm{TAC}(j)$ represents the AA from tasks 1 to j after training on task j, and Ave represents the average TAC from tasks 1 to T after the training of all tasks. TAC reflects the model's performance during the process of learning each task individually. Ave reflects the average performance across different tasks and helps assess the stability and generalization ability of the model.

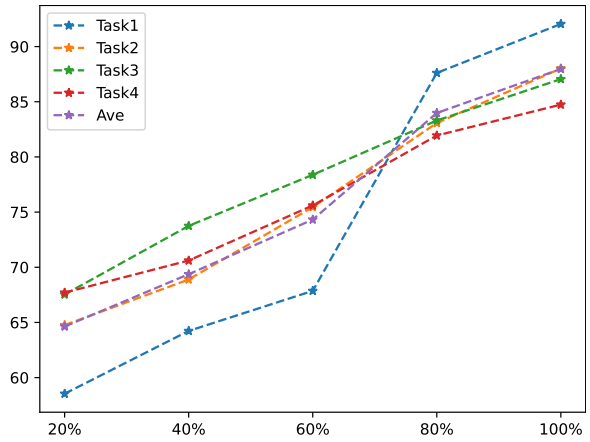


Fig. 5. Experiment results (%) of the different percentages of training samples on four HSI classification tasks.

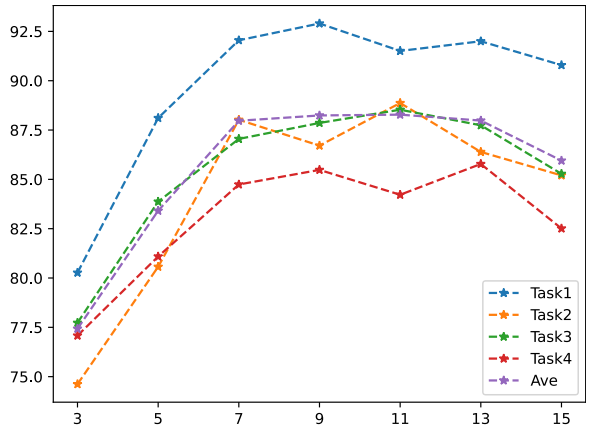


Fig. 6. Experiment results (%) of the different patch sizes on four HSI classification tasks.

C. Parameters Analysis

1) Training Samples Number: We select varying percentages of training samples from the provided training dataset, specifically 20%, 40%, 60%, 80%, and 100%. The results are reported in Fig. 5. As expected, the experimental results demonstrate a gradual improvement with an increasing percentage of training samples.

2) Patch Size: Generally, a larger patch size can capture more spatial information. However, excessively large patch sizes may result in information redundancy, thereby affecting classification performance. To investigate the impact of different patch sizes, we conduct experiments with different patch sizes of 3, 5, 7, 9, 11, 13, and 15. The experimental results are presented in Fig. 6. The experimental results indicate

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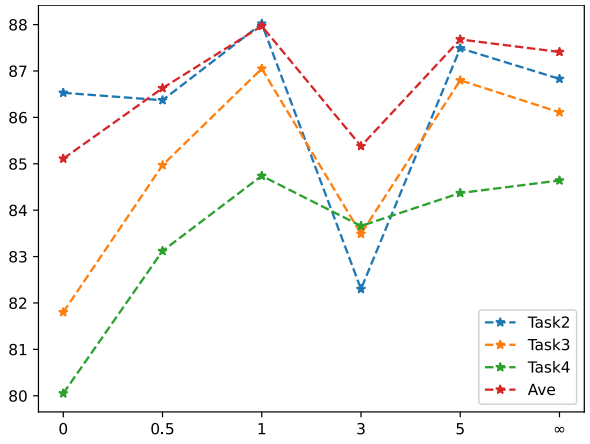


Fig. 7. Experiment results (%) of the different σ on tasks 2–4.

TABLE II
TAC (%) OF THE DIFFERENT EXEMPLAR SELECTION METHODS ON FOUR HSI CLASSIFICATION TASKS

Method	Random	Distance [53]	Entropy [53]	Herding [23]	MMCS
Task1	88.76±3.39	88.76±3.39	88.76±3.39	88.76±3.39	88.76±3.39
Task2	83.32±3.32	74.32±3.59	78.97±2.45	82.03±2.09	**86.15±1.26**
Task3	82.68±2.10	73.20±2.63	76.06±3.39	82.98±1.45	**85.03±1.98**
Task4	80.45±1.78	70.89±1.88	72.64±2.40	82.15±0.92	**83.67±0.89**
Ave	83.80±2.39	76.79±2.38	79.11±2.41	83.98±1.36	**85.90±1.24**

TABLE III
TAC (%) OF THE COMBINATION OF DIFFERENT COMPONENTS ON FOUR HSI CLASSIFICATION TASKS

Method	Task1	Task2	Task3	Task4	Ave
Baseline	88.76±3.39	47.97±3.00	32.49±0.22	21.62±0.34	47.71±1.42
Baseline+MMCS	88.76±3.39	81.28±4.26	82.47±2.46	81.17±0.91	83.42±2.26
Baseline+MMCS+CSSKD	88.76±3.39	**86.15±1.26**	**85.03±1.98**	**83.67±0.89**	**85.90±1.24**

that classification accuracy gradually increases as the patch size increases from 3 to 7. When the patch size reaches 7, the classification accuracy tends to stabilize. However, further increasing the patch size to 13 results in a decline in classification accuracy. In this study, we set the patch size to 7.

3) Standard Deviation (σ) in SSGD: We investigate the impact of the standard deviation (σ) in SSGD. We set σ to different values, including 0, 0.5, 1, 3, 5, and ∞. Specifically, when $\sigma = 0$, only the central pixel contributes to the calculation of SSGD, while the weights of surrounding pixels are zero. Conversely, when $\sigma = \infty$, all pixels in the patch are assigned equal weights. An increase in σ indicates that the weights of the edge pixels in the patch gradually approach the weight of the central pixel, and vice versa. As illustrated in Fig. 7, as σ increases from 0 to ∞, the overall classification performance of the model initially improves, followed by a decline. The highest performance in four classification tasks is observed at $\sigma = 1$, achieving the highest TAC in tasks 2–4 with 88.02%, 87.05%, and 84.74%, respectively. The classification performance displays instability at $\sigma = 3$. Our hypothesis, which posits that the center pixel assumes a more pivotal role than its surrounding pixels, was corroborated by the experimental results. In this article, we set σ to 1.

D. Comparison With Other Exemplar Selection Methods

To assess the performance of MMCS, we conduct a comparative analysis with other exemplar selection methods, including Herding [23], Entropy, Distance, and Random [53]. The experimental results, obtained by averaging the results from five independent runs, are presented in Table II. The results indicate that the Distance method exhibits the lowest performance among all evaluated methods. This is attributed to the fact that selecting only a limited number of exemplars close to the decision boundary primarily captures the complexity of the data, but consequently lacks overall representativeness. Both Random and Herding methods outperform Entropy, with Herding surpassing Random in tasks 3 and 4, while Random outperforms Herding in task 2. MMCS achieves the best performance. MMCS outperforms Herding by 4.12%, 2.05%, and 1.52% on tasks 2–4, as Herding selects only representative samples as exemplars without accounting for diverse and complex samples. The experimental results demonstrate that our proposed MMCS effectively selects the most representative or diverse samples, thereby identifying the optimal exemplars for replay.

E. Ablation Experiments

To assess the effectiveness of each module within our proposed lifelong learning framework, we conduct ablation studies on four HSI classification tasks. We use HyperKD without the MMCS and CSSKD modules as a baseline. Our method primarily consists of two components: MMCS and CSSKD. The experimental results, obtained by averaging the results from five independent runs, are summarized in Table III, revealing the following observations.

1) The incorporation of the MMCS module improves the baseline results by 33.31%, 49.98%, and 59.55% on tasks 2–4, respectively. The performance improvement can be attributed to the MMCS module, which selects the most representative or diverse samples as exemplars and stores them in a memory buffer to ensure that at least some information about the data distribution of previous classes can be incorporated into the training for the new task.
2) Notably, the combination of MMCS and CSSKD achieves the best performance. Compared to the exclusive use of MMCS, this combined method exhibits improvements of 4.87%, 2.56%, and 2.50% on tasks 2–4, respectively. This improvement can be attributed to the CSSKD's ability to transfer the spectral–spatial feature distribution and the prediction logit distribution from the previous network to the current network, thereby retaining more knowledge and mitigating forgetting. The experimental results clearly demonstrate that the two proposed modules in our lifelong learning framework have significant contributions to lifelong learning for HSI classification.

F. Comparison of Different Lifelong Learning Methods

We conduct experiments to compare the classification performance of several lifelong learning approaches for HSI.

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TABLE IV
LIFELONG LEARNING PERFORMANCE (%) OF DIFFERENT METHODS ON FOUR HSI CLASSIFICATION TASKS

Task	Dataset		Finetuning	EWC [26]	LWF [31]	FCS [35]	iCaRL [23]	CSSFD [49]	CSSFD* [49]	A-MAS* [25]	Ours
Task1	Pavia	OA	**88.76±3.39**	86.90±4.17	**88.76±3.39**	83.44±1.03	85.40±3.72	84.69±0.58	84.69±0.58	85.77±4.26	**88.76±3.39**
		TAC	**88.76±3.39**	86.90±4.17	**88.76±3.39**	83.44±1.03	85.40±3.72	84.69±0.58	84.69±0.58	85.77±4.26	**88.76±3.39**
Task2	Pavia	OA	2.65±5.15	3.54±6.16	3.29±4.52	3.17±1.16	73.54±1.41	**80.17±2.77**	45.12±5.67	77.49±2.09	79.61±1.91
	Indian Pines	OA	**93.29±1.08**	89.93±4.95	86.75±4.49	61.26±2.79	91.02±1.19	60.97±6.49	70.30±23.38	90.96±0.99	92.68±0.82
		TAC	47.97±3.00	46.74±4.58	45.02±2.96	32.22±1.42	82.28±0.96	70.57±3.48	57.71±12.41	84.23±1.15	**86.15±1.26**
Task3	Pavia	OA	0.00±0.00	0.00±0.00	0.54±1.09	9.63±17.90	68.74±1.85	**79.94±4.00**	39.25±10.26	57.65±20.07	71.07±5.44
	Indian Pines	OA	0.00±0.00	0.03±0.05	0.09±0.17	9.99±4.77	85.76±0.76	71.99±5.75	35.76±4.97	82.47±2.23	**86.98±1.23**
	Salinas	OA	**97.15±0.20**	97.00±0.35	97.12±0.36	85.14±2.05	96.57±0.44	89.26±3.75	85.93±7.52	95.49±0.62	97.03±0.33
		TAC	32.38±0.07	32.34±0.11	32.58±0.42	34.92±6.21	83.69±0.47	80.39±2.73	53.65±4.16	78.54±7.42	**85.03±1.98**
Task4	Pavia	OA	0.00±0.00	0.00±0.00	0.00±0.00	0.00±0.00	66.84±1.60	**78.00±2.39**	37.05±11.67	68.70±6.24	73.26±3.40
	Indian Pines	OA	0.06±0.12	0.00±0.00	0.00±0.00	1.08±1.06	83.30±1.03	76.71±3.38	32.96±1.68	80.87±2.35	**83.55±1.58**
	Salinas	OA	0.56±1.13	0.32±0.64	0.94±1.88	10.73±10.07	**93.31±0.53**	91.53±2.80	55.80±2.10	88.29±2.17	93.15±0.90
	Houston	OA	**85.87±1.15**	82.07±6.83	79.79±6.36	68.62±1.78	81.37±2.11	74.45±0.85	73.29±2.49	82.18±0.84	84.73±1.06
		TAC	21.62±0.34	20.60±1.74	20.18±1.58	20.11±2.75	81.20±0.73	80.17±0.86	49.77±3.55	80.01±1.18	**83.67±0.89**
		Ave	47.68±1.46	46.64±1.51	46.64±1.42	42.67±1.56	83.14±1.38	78.96±0.98	61.45±2.55	82.13±2.26	**85.90±1.24**

In addition to our method proposed in this study, we have also compared conventional lifelong learning methods, including Finetuning, LWF [31], EWC [26], FCS [35], iCaRL [23], CSSFD [49], and A-MAS* [25]. In FCS, we use the nearest-mean-of-exemplars classification strategy to predict labels. We utilize our MMCS strategy, which retains a small fraction of samples, as an alternative to retaining all samples in the CSSFD method, subsequently designating this refined method as CSSFD*. A-MAS* is an auxiliary network lifelong learning method [25] based on MAS [27], which integrates our MMCS module to enable exemplar replay.

In task 1, the Pavia dataset serves as both the training and testing set to evaluate the model's ability to learn new classes. In task 2, we employ the Indian Pines dataset as the new data to assess the model's learning ability for new classes. Concurrently, the Pavia dataset serves as the seen data, primarily used to evaluate the model's ability to retain prior knowledge. In task 3, the Salinas dataset serves as the new data to test the model's learning ability, while the Pavia and Indian pines datasets jointly serve as prior data, evaluating the model's performance in mitigating the forgetting of previous knowledge. Task 4 broadens the scope further by utilizing the Houston dataset as the new data to evaluate the model's learning ability. Additionally, the Pavia, Indian pines, and Salinas datasets collectively serve as prior data, comprehensively assessing the model's effectiveness in retaining previous knowledge.

The OA results of the comparative experiments for different lifelong learning methods in four HSI classification tasks are summarized in Table IV. The results are obtained by averaging the results from five independent runs. From the experimental results, it can be observed that the four methods of finetuning, EWC, LWF, and FCS generally struggle to retain previously acquired knowledge effectively. Specifically, in task 3, the OA of finetuning, EWC, and LWF on the Pavia and Indian Pines datasets drops to nearly 0%, while the OA of FCS is only 9.63% in the Pavia dataset and 9.99% in the Indian Pines dataset. In task 4, the OA of the four methods drops to 0% in the Pavia dataset. The OA of Finetuning, EWC, and LWF in the Indian Pines and Salinas datasets drops to nearly 0%, while FCS achieves an OA of 1.08% in the Indian Pines dataset and an OA of 10.73% in the Salinas dataset. The primary reason for this phenomenon is that the four lifelong learning methods lack mechanisms to store or utilize data from previous tasks during training. When a new task is encountered, these methods can only rely on the current task's dataset for training, unable to access data from previous tasks. As a result, with the continuous accumulation of tasks, previously learned knowledge is gradually forgotten, leading to a significant decline in classification performance. Although the FCS method establishes a prototype for each previously learned class, using it as a reference from historical data to train the model for subsequent tasks, these prototypes are not yet effective in retaining the previously acquired knowledge in long-sequence tasks. The CSSFD method, although not the most optimal in overall classification capability, demonstrates excellent performance in retaining learned knowledge. In task 2, the OA on the Pavia dataset decreases by only 4.52% compared to task 1. In task 3, the OA on the Pavia dataset decreases by only 0.23% compared to task 2. However, it unexpectedly increases by 11.02% on the Indian Pines dataset. In task 4, the OA on the Pavia dataset decreases by 1.94% compared to task 3. However, it unexpectedly increases by 4.72% on the Indian Pines dataset and by 2.27% on the Salinas dataset. This phenomenon is mainly attributed to the CSSFD method, which retains all training data from previous tasks whenever new tasks are introduced, thereby assisting the training of new tasks. Essentially, the CSSFD method is a type of joint training method. As shown in Table IV, our proposed lifelong learning method achieves superior classification performance in four HSI classification tasks, closely followed by iCaRL and A-MAS*. In task 2, our method achieves a TAC of 86.15%, outperforming the A-MAS* method by 1.92% and the iCaRL method by 3.87%. For task 3, the TAC remains high at 85.03%, surpassing the iCaRL method by 1.34% and the CSSFD method by 4.64%. In task 4, our method maintains a stable TAC of 83.67%, outperforming the A-MAS* method by 3.66% and the iCaRL method by 2.47%. Throughout the four tasks, our method achieves an Ave of 85.90%, exceeding the A-MAS* method by 3.77% and the iCaRL method by 2.76%. These results conclusively demonstrate that our method effectively learns

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TABLE V
CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH CLASS ON FOUR HSI CLASSIFICATION TASKS

Pavia					Indian pines				Salinas			Houston	
Class	Task1	Task2	Task3	Task4	Class	Task2	Task3	Task4	Class	Task3	Task4	Class	Task4
0	92.20±2.30	76.45±10.75	54.04±25.97	63.53±7.74	9	88.68±1.39	79.64±4.24	76.21±6.96	25	100.00±0.00	100.00±0.00	41	82.15±0.18
1	88.53±8.76	80.32±4.06	81.05±3.45	76.12±5.10	10	96.71±1.26	88.55±3.61	88.04±4.00	26	100.00±0.00	99.99±0.02	42	98.27±2.17
2	80.00±6.17	72.19±9.63	60.73±23.08	73.67±14.33	11	100.00±0.00	92.61±4.03	87.17±3.99	27	100.00±0.00	92.79±11.84	43	96.75±2.43
3	83.34±6.65	74.16±9.30	66.68±22.97	64.29±16.15	12	94.54±0.91	84.61±6.16	74.90±4.41	28	99.97±0.07	99.65±0.35	44	96.78±0.99
4	99.93±0.10	99.12±0.69	94.21±1.92	97.02±0.90	13	99.14±0.62	97.82±1.62	98.59±1.20	29	99.75±0.22	98.64±1.77	45	99.92±0.11
5	87.72±6.55	78.79±6.23	54.65±23.84	71.54±19.62	14	100.00±0.00	100.00±0.00	99.68±0.27	30	100.00±0.00	99.99±0.01	46	96.50±1.88
6	84.83±15.05	77.23±22.31	64.89±33.07	83.53±6.18	15	86.86±0.86	80.37±3.98	82.53±3.64	31	99.99±0.02	99.89±0.08	47	88.73±4.06
7	89.96±5.10	82.89±7.56	70.45±21.26	69.81±13.46	16	88.31±2.02	83.28±4.11	74.25±8.76	32	93.83±0.91	78.93±7.26	48	73.03±7.57
8	96.70±1.04	91.80±1.40	90.42±4.03	95.65±2.76	17	89.57±2.13	84.22±4.31	81.45±4.55	33	99.99±0.01	100.00±0.00	49	83.97±1.71
					18	99.75±0.30	99.63±0.30	99.38±0.68	34	99.30±0.08	95.01±0.21	50	53.53±6.64
					19	97.93±0.27	94.18±2.96	94.21±3.08	35	99.77±0.13	99.35±0.84	51	68.96±8.08
					20	98.06±1.93	87.27±4.70	81.76±4.84	36	99.91±0.11	98.99±1.73	52	85.30±2.84
					21	100.00±0.00	98.22±2.59	99.11±1.09	37	100.00±0.00	100.00±0.00	53	82.46±4.23
					22	97.95±1.03	98.97±1.26	100.00±0.00	38	99.84±0.12	99.66±0.52	54	99.60±0.51
					23	100.00±0.00	100.00±0.00	100.00±0.00	39	88.72±1.90	88.63±5.87	55	97.25±2.90
					24	100.00±0.00	100.00±0.00	100.00±0.00	40	99.99±0.02	99.93±0.10		
OA	88.76±3.39	79.61±1.91	71.07±5.44	73.26±3.40		92.68±0.82	86.98±1.23	83.55±1.58		97.03±0.33	93.15±0.90		84.73±1.06
AA	89.24±2.42	81.44±2.84	70.79±6.93	77.24±3.65		96.10±0.49	91.84±0.72	89.83±0.62		98.82±0.14	96.97±0.74		86.88±0.99
Kappa	84.77±4.58	73.17±2.59	61.48±7.10	64.95±4.59		91.62±0.93	85.10±1.38	81.27±1.72		96.67±0.37	92.36±0.99		83.41±1.15

TABLE VI
COMPLEXITY AND RUNNING TIME OF DIFFERENT METHODS ON FOUR HSI CLASSIFICATION TASKS

	Task	Finetuning	EWC [26]	LWF [31]	FCS [35]	iCaRL [23]	CSSFD [49]	CSSFD* [49]	A-MAS* [25]	Ours
Params (K)	Task1	488.77	488.77	488.77	549.91	488.77	7336.94	7336.94	488.77	488.77
	Task2	490.71	490.71	490.71	567.82	490.71	7345.15	7345.15	490.71	490.71
	Task3	492.64	492.64	492.64	575.56	492.64	7353.35	7353.35	492.64	492.64
	Task4	494.46	494.46	494.46	581.01	494.46	7361.56	7361.56	494.46	494.46
Time Consumer (ms)	Task1	381.73	389.44	394.92	1611.23	372.54	988.72	1006.76	378.37	378.77
	Task2	72.74	136.02	109.61	439.63	107.09	1864.36	296.06	193.73	105.21
	Task3	317.11	553.63	421.07	1625.43	560.63	3098.48	1324.63	836.92	464.15
	Task4	275.80	500.99	396.12	1486.13	613.89	4251.62	1164.04	786.74	441.49

new categories while maintaining classification performance for previous categories, significantly mitigating catastrophic forgetting and optimizing overall classification performance.

Specifically, Table V demonstrates the classification performance of our proposed lifelong learning method for each class across four HSI datasets. Across the four tasks, our method demonstrates the ability to not only effectively learn new knowledge in the new task, but also retain the knowledge acquired from previous tasks.

Furthermore, we conduct experimental analyses to evaluate the model parameter count and the training time consumption of different methods. As shown in Table VI, compared with all other methods, our method exhibits the lowest level of model parameter count. Meanwhile, the CSSFD method has a significantly higher model parameter count due to its specialized backbone network. In terms of training time consumption, although our method is not optimal per training epoch, it is very close to the optimal level. Compared with methods with similar classification performance, our method achieves an optimal balance between classification accuracy and efficiency, while significantly reducing both the training time and the complexity of the model.

Based on the comprehensive analysis of the experimental results, we draw the following comparisons.

1) LWF, EWC, and FCS are exemplar-free lifelong learning methods, which align more closely with practical application requirements. However, these methods have not been effective in mitigating catastrophic forgetting.
2) The iCaRL method introduces the Herding strategy to select some samples as exemplars for replay and learns a prototype for each class. However, it overly focuses on representativeness, neglecting diversity, which restricts the expansion of decision boundaries.
3) The CSSFD method has designed a specialized backbone, which not only reduces generalizability but also compromises its computational efficiency and classification performance.
4) A-MAS*, an innovation based on the MAS [27], designs an auxiliary network to enhance the learning ability for new tasks. However, while this design increases the learning ability for new tasks, it also inhibits the classification ability for old tasks to a certain degree.
5) Our method, HyperKD, achieves significant results in experiments, but it still has certain limitations. First, it relies on the exemplar replay strategy, which may not be feasible in some real-world scenarios, thus limiting its application scope. Second, as the number of tasks increases, so do the classification heads, leading to model bloat and decreasing classification efficiency. A potential solution is to explore unsupervised lifelong learning methods.

G. Visualize Results

In lifelong learning, as the number of tasks increases, the necessary classification categories gradually expand. For example, in our experiment, task 4 requires distinguishing between 56 categories. Using 56 different colors to visualize the classification results leads to poor visualization due to decreased color contrast. To enhance visualization, we introduce an offset for visualizing each task, equivalent to the

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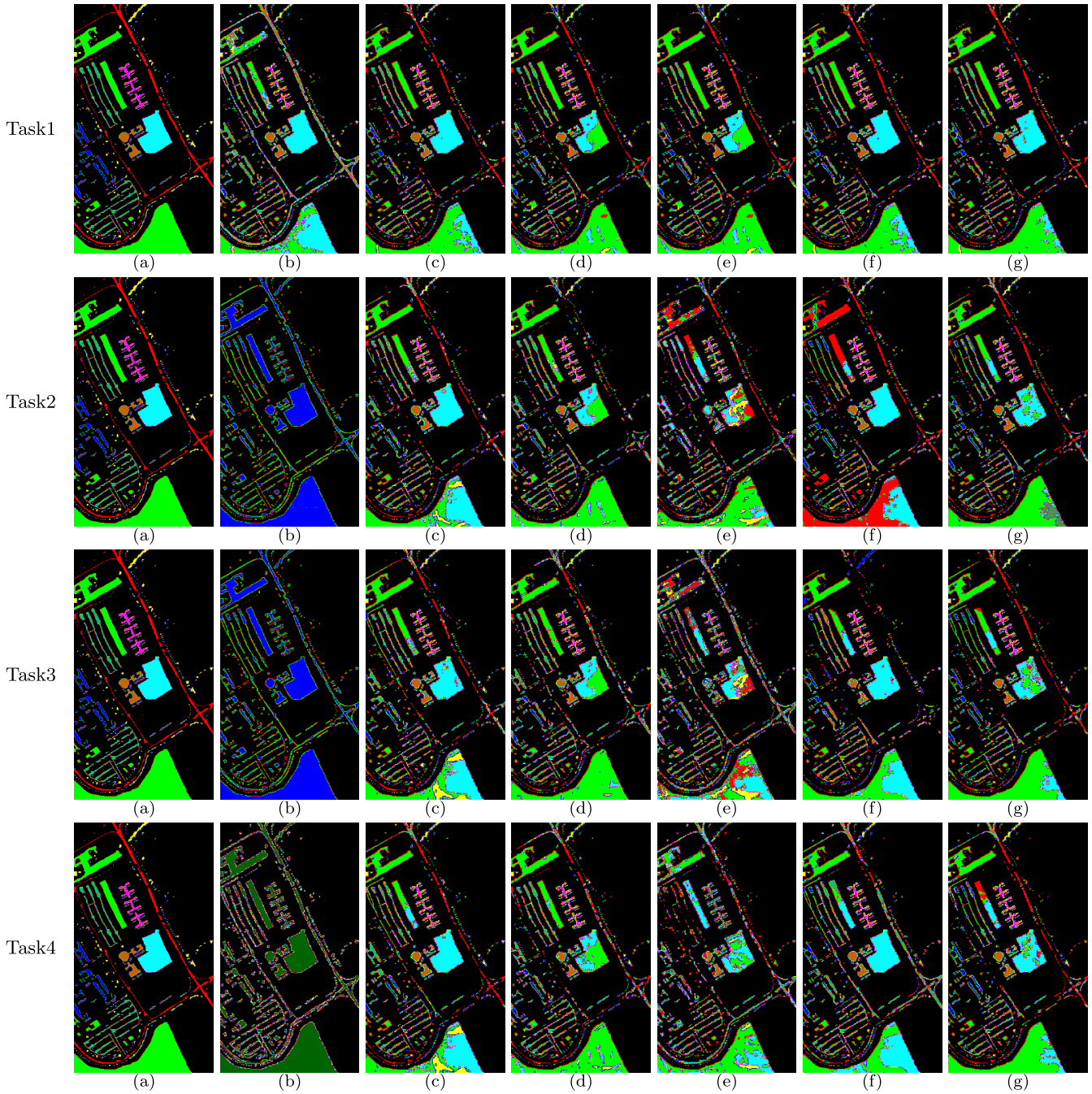


Fig. 8. Classification maps for tasks 1–4 on the Pavia dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours.

cumulative number of categories from previous tasks. We then subtract this offset from the prediction labels to ensure that the labels for each task start from 0. Additionally, we merge misclassified categories across tasks to minimize the number of incorrect categories. This processing approach effectively reduces the total number of categories, making the visualization results more concise and clear. The classification maps of several comparison methods are presented in Figs. 8–11, corresponding to the Pavia, Indian Pines, Salinas, and Houston, respectively.

Fig. 8 presents the classification maps of the Pavia dataset in tasks 1–4. In task 1, the dataset serves as the training dataset, whereas in tasks 2–4, it solely serves as the testing dataset. The following observations are made: In task 1, FCS, iCARL, and A-MAS* introduce noise in the green region located at the bottom of the images. The CSSFD and CSSFD* methods introduce significantly more noise in the middle region of the images. In task 2, CSSFD* and A-MAS* demonstrate significant misclassifications throughout the images, as indicated by the red-labeled regions. In tasks 3 and 4, the noise introduced by CSSFD, CSSFD*, and iCARL in the bottom green region of the images increases further. The classification capabilities of FCS are completely lost in tasks 2–4 due to its inability to retain previously acquired knowledge. Intuitively, our method is the cleanest and the closest to the ground-truth map.

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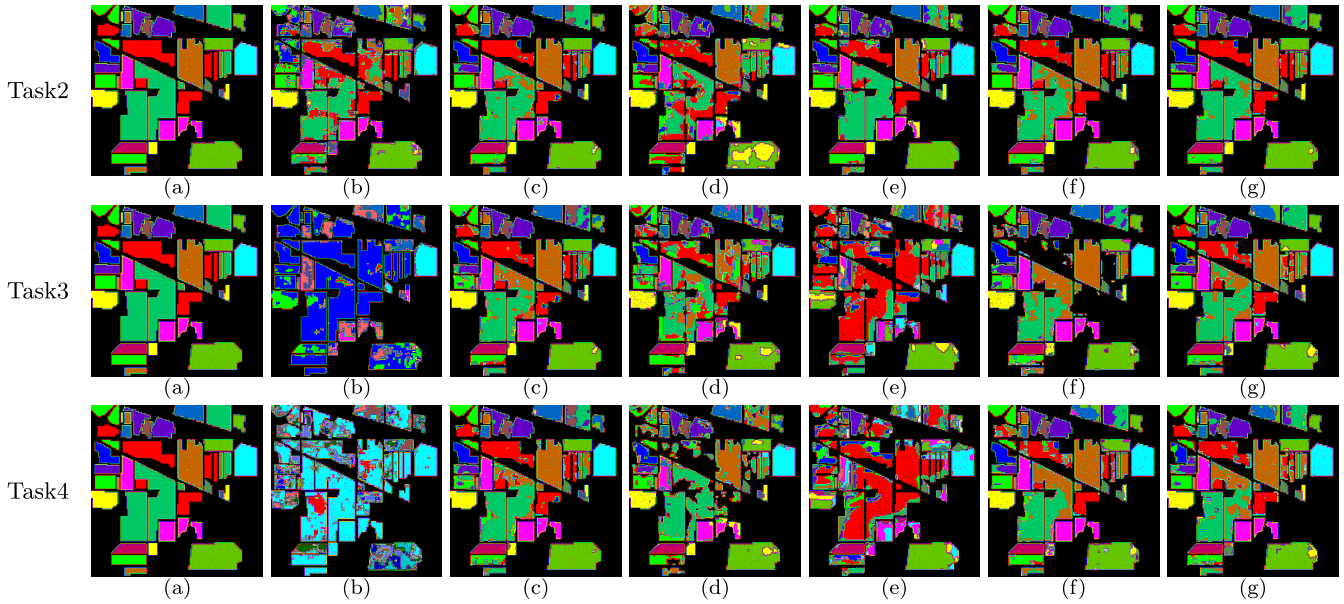


Fig. 9. Classification maps for tasks 2–4 on the Indian Pines dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours.

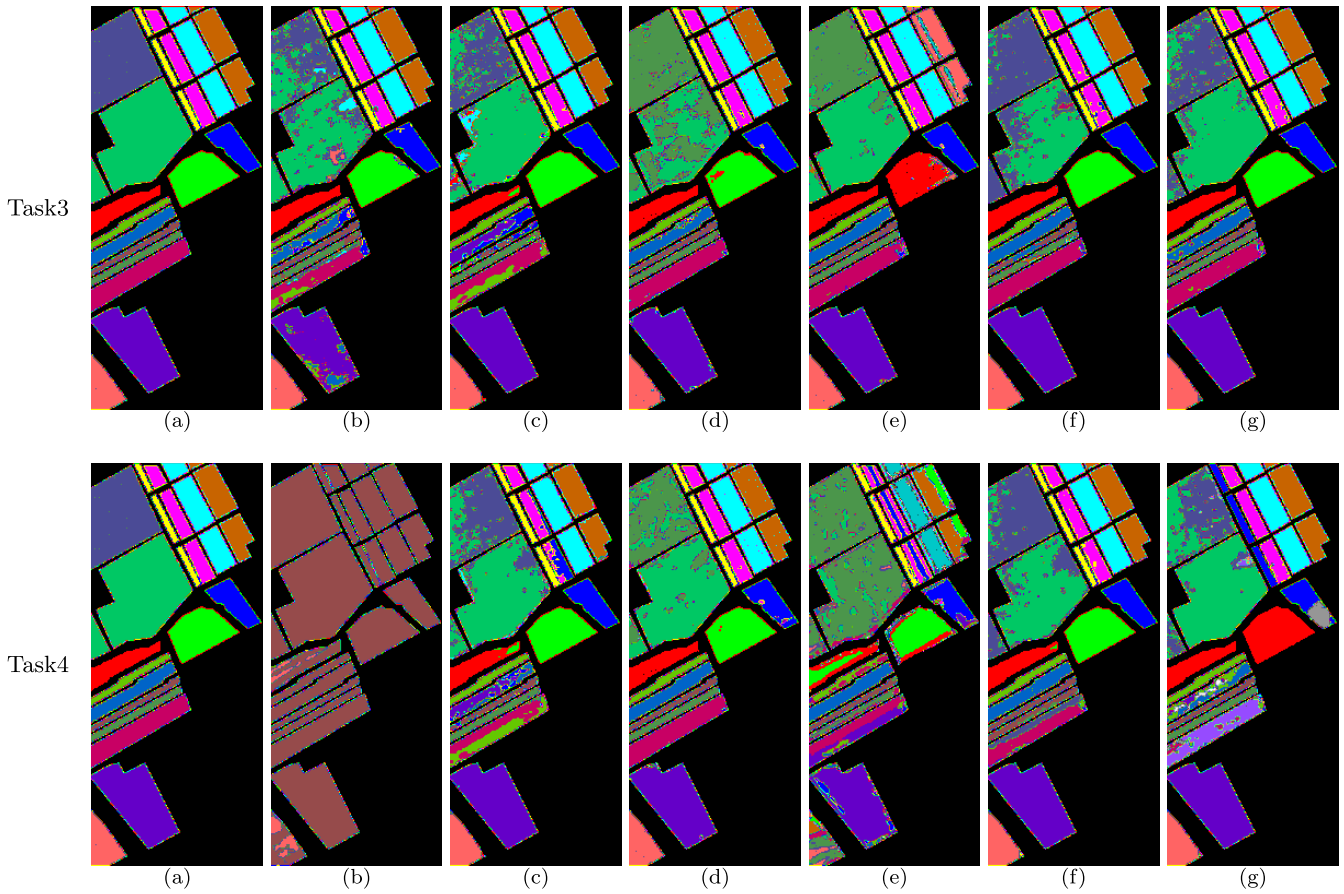


Fig. 10. Classification maps for tasks 3 and 4 on the Salinas dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours.

The classification maps of the Indian Pines dataset are shown in Fig. 9. For task 2, the dataset serves as the training dataset, whereas in tasks 3 and 4, it solely serves as the testing dataset. In task 2, CSSFD incorrectly classifies certain green areas as red, introducing significant noise. In tasks 3 and 4, the CSSFD* method introduces significantly more noise throughout the image, as indicated by the red-labeled regions. The classification capabilities of FCS are completely lost in tasks 3 and 4.

The classification maps for the Salinas dataset are presented in Fig. 10. In task 3, this dataset serves as the training dataset, whereas in task 4, it solely serves for testing. During task 3,

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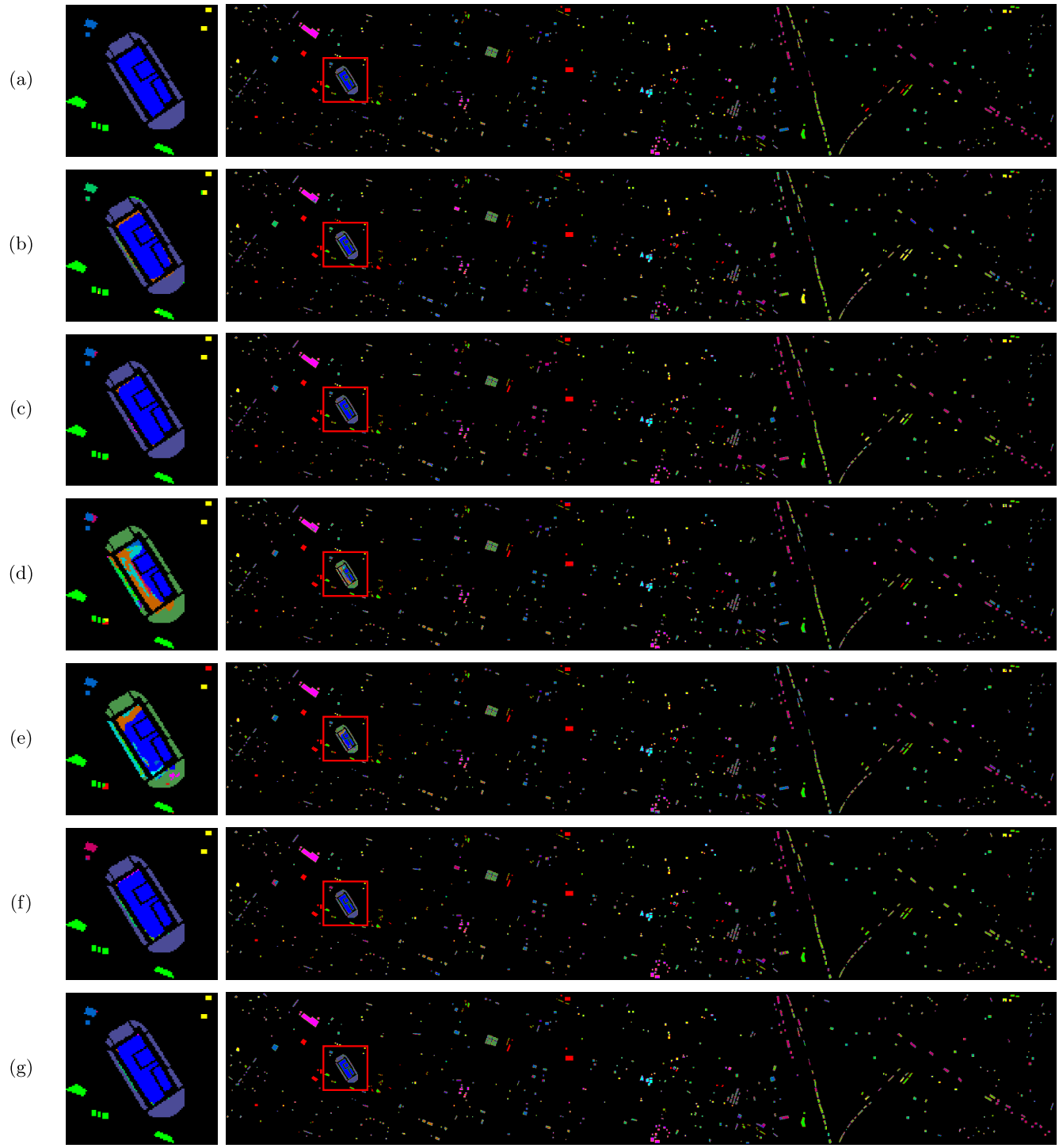


Fig. 11. Classification maps for task 4 on the Houston dataset. The left image provides an enlarged view of the area indicated by the red box in the right image. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours.

the CSSFD* method incorrectly classifies a green region in the middle as red, introducing significant noise. In task 4, intuitively, iCaRL, A-MAS*, and our method exhibit the highest overall classification accuracy, closely approximating the ground truth.

The Houston dataset is exclusively used as the training dataset for task 4, with the classification maps presented in Fig. 11. For a clearer visualization of the classification results, we focus on the red-boxed areas in Fig. 11, with corresponding zoomed-in images presented on the left. Upon observation, CSSFD* and CSSFD exhibit significant errors at the edge of the playground and introduce notable noise within the playground area. A-MAS* also introduces some noise at the edge of the playground and in the top-left corner of the image, as indicated by the pink-marked areas. Visually, our method and iCaRL are closest to the ground truth, with the least noise.

To present the details of specific key areas in Figs. 8–10 more clearly, we zoomed-in view on a key region within these

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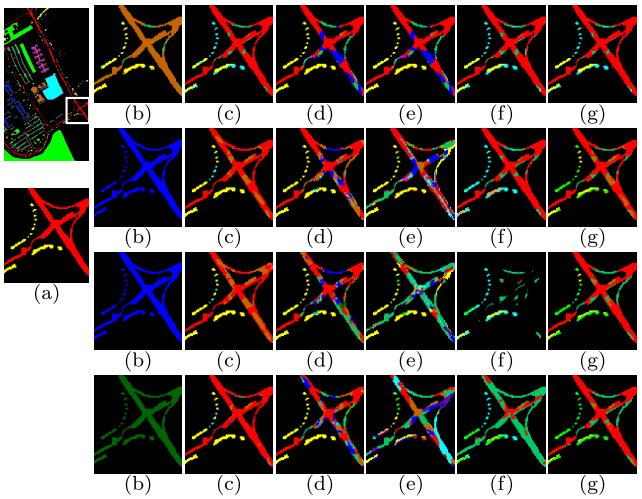


Fig. 12. Local enlarged classification maps for tasks 1–4 on the Pavia dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours. Row 1 corresponds to task 1, row 2 corresponds to task 2, row 3 corresponds to task 3, and row 4 corresponds to task 4.

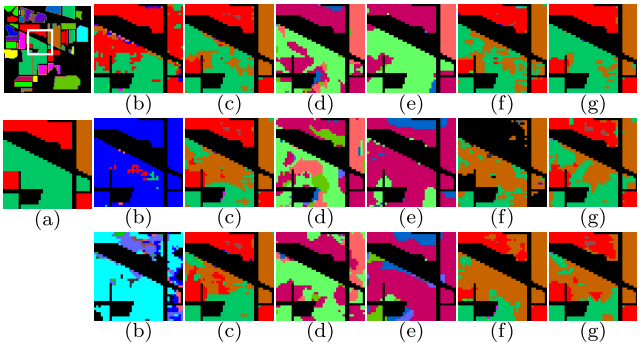


Fig. 13. Local enlarged classification maps for tasks 2–4 on the Indian Pines dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours. Row 1 corresponds to task 2, row 2 corresponds to task 3, and row 3 corresponds to task 4.

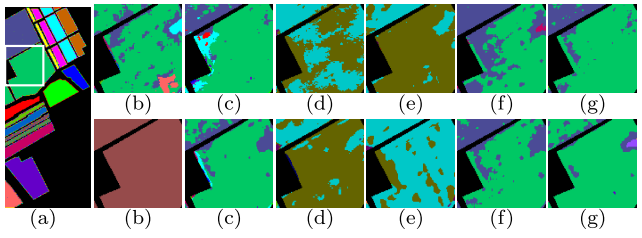


Fig. 14. Local enlarged classification maps for tasks 3 and 4 on the Salinas dataset. (a) Ground truth. (b) FCS. (c) iCaRL. (d) CSSFD. (e) CSSFD*. (f) A-MAS*. (g) Ours. Row 1 corresponds to task 3, and row 2 corresponds to task 4.

figures and present corresponding locally enlarged classification maps in Figs. 12–14. It can be clearly seen from these figures that our method is closer to the ground truth and has less noise.

V. Conclusion

In this article, we propose a novel lifelong learning framework for HSI classification, termed HyperKD. This method comprises an MMCS module that establishes an SSGD to assess sample similarity and cross-selects the most representative or diverse samples as exemplars for replay in future tasks. Additionally, it incorporates a CSSKD module, designed to transfer the spectral–spatial feature distribution from the previous network to the current network, without directly measuring the similarity between these distributions. Extensive experimental results and analysis demonstrate that our proposed method achieves optimal performance for HSI classification across four datasets. Our method exhibits the capability to not only efficiently learn new knowledge from new datasets in each task but also retain the knowledge acquired from previous tasks.

Currently, one of the main reasons why lifelong learning can achieve relatively good performance is the adoption of the exemplar replay strategy, which involves storing a portion of samples as exemplars from previous tasks for subsequent use. In future work, we will focus on two main areas: first, improving the performance of lifelong learning without exemplar replay strategies to achieve parity with exemplar replay methods; second, developing an unsupervised lifelong learning method tailored to practical application scenarios.

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Zhenlin Li received the bachelor's degree from the College of Physical Science and Technology, Sichuan University, Chengdu, China, in 2010, the master's degree in communication and information systems from the University of Chinese Academy of Sciences, Beijing, China, in 2016. He is currently pursuing the Ph.D. degree in control science and engineering with the College of Electrical and Information Engineering, Hunan University, Changsha, China.

His research interests include hyperspectral remote sensing image processing and deep learning.

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Shaobo Xia received the bachelor's degree in geodesy and geomatics from the School of Geodesy and Geomatics, Wuhan University, Wuhan, China, in 2013, the master's degree in cartography and geographic information systems from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2016, and the Ph.D. degree in geomatics from the University of Calgary, Calgary, AB, Canada, in 2020.

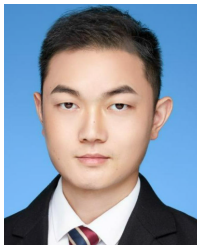
He is an Assistant Professor with the Department of Geomatics Engineering, Changsha University of Science and Technology, Changsha, China. His research interests include point cloud processing and remote sensing.



Jun Yue received the B.Eng. degree in geodesy from Wuhan University, Wuhan, China, in 2013, and the Ph.D. degree in GIS from Peking University, Beijing, China, in 2018.

He is currently an Associate Professor with the School of Automation, Central South University, Changsha, China. His research interests include satellite image understanding, pattern recognition, and few-shot learning.

Dr. Yue serves as a reviewer for IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING, *Information Fusion*, and *Information Sciences*.



Leyuan Fang (Senior Member, IEEE) received the Ph.D. degree from the College of Electrical and Information Engineering, Hunan University, Changsha, China, in 2015.

From August 2016 to September 2017, he was a Post-Doctoral Researcher with the Department of Biomedical Engineering, Duke University, Durham, NC, USA. He is currently a Professor with the College of Electrical and Information Engineering, Hunan University. His research interests include sparse representation and multiresolution analysis in remote sensing and medical image processing.

Dr. Fang was a recipient of the 2nd-grade National Award at the Nature and Science Progress of China in 2019. He is an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and *Neurocomputing*.

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